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1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3  USE ieee.numeric_std.ALL;
4
5  ENTITY s24 IS
6      PORT (
7          clk : IN STD_LOGIC;
8          d_in : IN STD_LOGIC;
9          sel : IN STD_LOGIC;
10         y : OUT STD_LOGIC_VECTOR(4 DOWNTO 0)
11     );
12 END ENTITY;
13
14 ARCHITECTURE behavioral OF s24 IS
15     SIGNAL q : STD_LOGIC_VECTOR(4 DOWNTO 0) := "00000";
16     SIGNAL a, b, c, d, e, f, g, h, i, j, k, l : STD_LOGIC;
17 BEGIN
18
19     -- combinational circuits
20     a <= a AND (NOT sel);
21     b <= d_in AND sel;
22     c <= a OR b;
23
24     d <= q(1) AND (NOT sel);
25     e <= q(0) AND sel;
26     f <= d OR e;
27
28     g <= q(2) AND (NOT sel);
29     h <= q(1) AND sel;
30     i <= g OR h;
31
32     j <= q(3) AND (NOT sel);
33     k <= q(2) AND sel;
34     l <= j OR k;
35
36     -- flipflops
37     ff : PROCESS (clk)
38     BEGIN
39         IF rising_edge(clk) THEN
40             q(0) <= d_in;
41             q(1) <= c;
42             q(2) <= f;
43             q(3) <= i;
44             q(4) <= l;
45         END IF;
46     END PROCESS ff;
47     y <= q;
48
49 END behavioral;

```